No.	Solution	Remarks
1(a)	$I = f^2 A^2 v k$ $k = \dfrac{I}{f^2 A^2 v}$ SI base units of k $= \dfrac{\text{W m}^{-2}}{\text{s}^{-2}\ \text{m}^2\ \text{m s}^{-1}} = \dfrac{\text{kg m}^2\ \text{s}^{-3}\ \text{m}^{-2}}{\text{s}^{-3}\ \text{m}^3} = \text{kg m}^{-3}$	[1] for correct units of frequency and amplitude [1] for correct base units of power [1] for correct answer
1(b)	$P = v^3 b$ $v = \sqrt[3]{\dfrac{P}{b}} = \sqrt[3]{\dfrac{84\times10^3}{0.56}} = 53.133\ \text{m s}^{-1}$ $P = v^3 b$ $v = \sqrt[3]{\dfrac{P}{b}}$ $\dfrac{\Delta v}{v} = \dfrac{1}{3}\dfrac{\Delta P}{P} + \dfrac{1}{3}\dfrac{\Delta b}{b}$ $\dfrac{\Delta v}{53.133} = \dfrac{1}{3}(0.05) + \dfrac{1}{3}(0.07)$ $\Delta v = 2.1253\ \text{m s}^{-1}$ $\Delta v \approx 2\ \text{m s}^{-1}$	[1] for correct value of v [1] for correct substitution [1] for correct answer to 1 s.f.
2(a)(i)	Applying Newton's second law	[1] for correct